

Resilient Control and Leader Switching for Autonomous Vehicle Platoons Under DoS Attacks

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Abstract—Autonomous vehicle platoons depend on directed communication for coordination, but are highly susceptible to Denial-of-Service (DoS) attacks that undermine stability and cooperation. This work addresses the challenging case in which both the leader and a follower are simultaneously subjected to DoS attacks. A resilient vehicular control strategy is proposed to detect, mitigate, and recover from such adversarial conditions. Attack detection is achieved through a divergence-based scheme that employs dual incremental timers to estimate the time-varying delay induced by the attack, using reference trajectories. Once the delay is quantified, a resilient controller is activated to restore consensus, regulate attacked vehicle dynamics, and ensure safety constraints. To preserve network functionality during prolonged disruptions, a switching mechanism reassigns the compromised leader as a follower and designates a new leader, thereby maintaining hierarchical structure and continuous guidance. The new leader is selected using a rule-based strategy that evaluates communication connectivity, spatial proximity, and motion similarity over short-horizon measurements relative to the former leader. Distributed control preserves consensus in the platoon under attack conditions. Simulation studies validate the proposed strategy, demonstrating its robustness and practical significance for secure platooning in intelligent transportation systems.

I. INTRODUCTION

Autonomous vehicular platoons consist of a group of intelligent vehicles that cooperate to reach a shared destination. Typically, a designated leader guides the movement of the group, while the remaining vehicles, referred to as followers, maintain a constant velocity and a prescribed inter-vehicle spacing relative to the leader [1]. The coordination among vehicles is established through a communication topology comprising vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) links [2], [3], enabling distributed control protocols to govern the dynamics of the platoon. Leader-follower consensus algorithms are commonly employed to ensure that each vehicle accurately tracks the reference position and velocity [4], [5].

However, the reliance on wireless communication channels renders connected vehicle systems vulnerable to cyber-physical threats, particularly Denial-of-Service (DoS) attacks [6]. Such attacks can overwhelm communication channels and introduce severe delays in data exchange, causing attacked vehicles to deviate from their intended trajectories and potentially compromising platoon safety [7].

**This work was supported in part by the U.S. National Science Foundation (NSF) through Grants CMMI-2145571 and OAC-2320972.

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While conventional network coordination schemes and standard information security techniques may be insufficient to protect collective vehicle behavior against adversarial disruptions [8], existing countermeasures, including intrusion detection systems (IDS) [9], H_∞ controllers, and robust consensus algorithms for intermittent communication [10], [11], are often limited to fixed topologies or undirected graph structures [12], [13].

Advanced strategies for directed switching topologies and higher-order systems remain underexplored [14]. Although artificial potential field (APF) methods have been applied to enhance formation control [15], real-world scenarios often involve dynamic attack patterns. For instance, the fixed-delay assumption in [16] does not reflect time-varying conditions, and the one-time interaction between detection and control limits adaptability. While [17], [18] addressed time-varying delays using adaptive mechanisms, distributed resilient control under DoS attacks targeting the leader remains an open problem.

In practical networked systems, DoS attacks often appear as packet loss, congestion, or buffering at the communication layer [19], [20]. However, from the viewpoint of distributed consensus control, these effects can be equivalently interpreted as time-varying delays in the received neighbor information. Such delay-based modeling has been widely adopted in cyber-physical systems literature, as it captures the dominant influence of DoS attacks on closed-loop stability while enabling tractable analysis and resilient control design.

In scenarios where both the leader and a follower are compromised, initial instability jeopardizes coordination. Even after leader recovery, an attacked follower continues to threaten consensus. To address this challenge, we propose a distributed resilient control framework that ensures platoon safety and consensus retention, even under simultaneous DoS attacks on the leader and a follower. The main contributions of this paper are summarized as follows:

1) Simultaneous leader and follower attack modeling: Unlike classical DoS-resilient platoon control studies that assume only follower vehicles are attacked or that the leader remains uncompromised, this work explicitly addresses the challenging scenario in which both the leader and a follower are simultaneously subjected to DoS attacks. This setting represents a critical yet largely unexplored vulnerability in leader-follower platoon architectures.

2) Attack-aware leader switching under DoS conditions: When the leader is compromised, a dynamic topology reconfiguration strategy is introduced in which the attacked

leader is demoted to a follower and a new leader is selected to preserve the spanning-tree structure. The leader reassignment is guided by a data-driven, rule-based multi-criteria decision strategy that evaluates short-horizon time-series measurements of vehicle positions and velocities in the vicinity of the compromised leader. Specifically, candidate vehicles are assessed based on (i) communication connectivity, (ii) spatial proximity to the former leader, and (iii) motion similarity inferred from recent velocity and position trajectories, enabling identification of a suitable replacement that minimizes transient disturbances. Unlike conventional resilient consensus frameworks with fixed leaders, the proposed distributed resilient vehicular control approach enables leader switching under active DoS attacks while maintaining platoon stability.

3) Integrated detection, topology switching, and resilient control: Distinct from existing works that treat attack detection, topology reconfiguration, and control design as separate problems, this paper integrates online DoS detection, delay-aware topology switching, and a resilient control strategy within a unified framework. The estimated delay is explicitly incorporated into the control decision-making process, enabling attack-severity-aware recovery and consensus restoration.

The remainder of this paper is organized as follows: Section II presents the platoon and DoS attack models. Section III details the detection algorithm, and Section IV introduces the resilient vehicular control strategy. Stability analysis is given in Section V, followed by simulation results in Section VI. Finally, Section VII concludes the work and outlines future directions.

II. MATHEMATICAL MODELLING

A. Platoon Model

Consider a homogeneous platoon consisting of N vehicles collaboratively engaged in tracking the position and velocity of a designated leader.

Definition 1: The inter-vehicle communication structure is represented by a directed graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ with a set of nodes $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$, a set of edges $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ and an adjacency matrix $\mathcal{A} = [a_{ij}]$.

In this formulation, the motion of each vehicle is modeled using a double-integrator dynamic system described by:

$$\dot{\xi}_i(t) = v_i(t), \quad \dot{v}_i(t) = u_i \quad (1)$$

where $\xi_i(t) \in \mathbb{R}$ and $v_i(t) \in \mathbb{R}$ denote the position and velocity of the i -th vehicle, respectively, and $u_i(t)$ represents the control input.

To attain leader-follower consensus, the following distributed control strategy is employed:

$$u_i(t) = -\sum_{j=1}^N a_{ij} \left((\hat{\xi}_i(t) - \hat{\xi}_j(t)) + \gamma(\hat{v}_i(t) - \hat{v}_j(t)) \right) \quad (2)$$

where $\gamma > 0$ is a tunable gain selected to guarantee the closed-loop stability of the platoon [13], $i \in \{1, 2, \dots, N\}$, and the associated tracking errors, representing deviations in

inter-vehicle spacing and relative velocity with respect to the leader, are defined as follows:

$$\begin{aligned} \hat{\xi}_i(t) &= \xi_i(t) - \xi_N(t) - d_{i,N} \\ \hat{v}_i(t) &= v_i(t) - v_N(t) \end{aligned} \quad (3)$$

where $d_{i,N}$ denotes the desired spacing between the i -th follower and the leader vehicle.

By defining the aggregated state vector of the platoon as $X(t) = [P^T(t), V^T(t)]^T$, the closed-loop system dynamics under nominal (attack-free) conditions are expressed as:

$$\dot{X}(t) = \Phi X(t), \quad \Phi = \begin{bmatrix} 0_{N \times N} & I_{N \times N} \\ -\mathcal{L} & -\gamma \mathcal{L} \end{bmatrix} \quad (4)$$

where the stacked vectors $P(t) = [\hat{\xi}_1(t), \hat{\xi}_2(t), \dots, \hat{\xi}_N(t)]^T$ and $V(t) = [\hat{v}_1(t), \hat{v}_2(t), \dots, \hat{v}_N(t)]^T$ represent the collective position and velocity states of all vehicles, respectively. Additionally, $\Phi \in \mathbb{R}^{2N \times 2N}$ denotes a structured system matrix, and $\mathcal{L} \in \mathbb{R}^{N \times N}$ is the Laplacian matrix associated with the underlying communication topology [13].

The closed-loop formulation in Eq. (4) is adopted as the nominal reference model for attack detection, which will be further detailed in Section III.

Assumption 1: In every considered scenario, the platoon's communication topology is structured to contain a spanning tree, with a leader positioned as the root node.

B. DoS Attack Model

A Denial-of-Service (DoS) attack poses a cyber threat that disrupts inter-vehicle communication by introducing unknown, time-varying delays. In a platoon of N vehicles, such delays are modeled as dynamic communication lags between vehicles. Let vehicle j^* be compromised by the DoS attack. The distributed control strategy for the remaining vehicles is then formulated as:

$$\begin{aligned} u_i(t) = & - \left[\sum_{j \in \mathcal{N}_i - j^*} a_{ij} \left((\hat{\xi}_i(t) - \hat{\xi}_j(t)) + \gamma(\hat{v}_i(t) - \hat{v}_j(t)) \right) \right] \\ & - a_{ij^*} \left((\hat{\xi}_i(t) - \hat{\xi}_{j^*}(t - \tau(t))) \right. \\ & \quad \left. + \gamma(\hat{v}_i(t) - \hat{v}_{j^*}(t - \tau(t))) \right) \\ & \pm \left[a_{ij^*} \left((\hat{\xi}_i(t) - \hat{\xi}_{j^*}(t)) + \gamma(\hat{v}_i(t) - \hat{v}_{j^*}(t)) \right) \right] \\ = & - \left[\sum_{j \in \mathcal{N}_i} a_{ij} \left((\hat{\xi}_i(t) - \hat{\xi}_j(t)) + \gamma(\hat{v}_i(t) - \hat{v}_j(t)) \right) \right] \\ & + a_{ij^*} \left((\hat{\xi}_{j^*}(t - \tau(t)) - \hat{\xi}_{j^*}(t)) \right. \\ & \quad \left. + \gamma(\hat{v}_{j^*}(t - \tau(t)) - \hat{v}_{j^*}(t)) \right) \end{aligned} \quad (5)$$

where $\mathcal{N}_i$ represents the set of neighboring vehicles for the i -th vehicle, and $\tau(t)$ denotes a time-varying delay induced by the DoS attack. The delay is assumed to be bounded above by a known constant $\mathbf{U}$, such that:

$$\begin{aligned} 0 &< \tau(t) < \mathbf{U}, \quad \mathbf{U} + \tau(t) < 2\mathbf{U} \\ 0 &< \tau(t) < \mathbf{U} < \mathbf{U} + \tau(t) < 2\mathbf{U} \\ \dot{\tau}(t) &< d < 1 \end{aligned} \quad (6)$$

where d represents the upper bound of the derivative of delay.

Remark 1: The condition $\dot{\tau}(t) < d < 1$ characterizes a slowly varying communication delay, whereas $1 < \dot{\tau}(t) < d$ indicates rapid fluctuations in delay. In this work, it is assumed that the delay evolves gradually over time, with its time derivative $\dot{\tau}(t)$ remaining bounded.

Remark 2: In the proposed framework, the induced communication delay is interpreted as a quantitative measure of DoS attack strength, where larger delays correspond to more severe congestion or blocking of communication channels. The adopted time-varying delay model captures worst-case network congestion effects and reflects the evolving intensity of DoS attacks. To ensure tractable analysis and meaningful control action, the delay is assumed to be bounded by an unknown bound, since unbounded delay corresponds to complete loss of connectivity under which no feedback-based control strategy can maintain stability or coordination. Consequently, bounding the delay represents a practical operating condition in which recovery and consensus restoration remain feasible, while preserving suitability for rigorous stability analysis and resilient control design.

The resulting dynamics derived from Eq. (5) can be expressed as follows:

$$\dot{X}(t) = \hat{\Phi}X(t) + \hat{\Phi}_1X(t - \tau(t)) \quad (7)$$

where $\hat{\Phi} \in \mathbb{R}^{2N \times 2N}$ determines the effect of the state and also $\hat{\Phi}_1 \in \mathbb{R}^{2N \times 2N}$ characterizes the influence of the delayed states, defined as:

$$\hat{\Phi} = \begin{bmatrix} 0_{N \times N} & I_{N \times N} \\ -\hat{\mathcal{L}}_{N \times N} & -\gamma \hat{\mathcal{L}}_{N \times N} \end{bmatrix}, \quad \hat{\Phi}_1 = \begin{bmatrix} 0_{N \times N} & 0_{N \times N} \\ -\mathcal{R} & -\gamma \mathcal{R} \end{bmatrix} \quad (8)$$

where $\mathcal{R} = [r_{ij}] \in \mathbb{R}^{N \times N}$ is considered as $\mathcal{R} = \mathcal{L} - \hat{\mathcal{L}}$.

It is assumed that a DoS attack targets both the leader and one of the follower vehicles in the platoon, with the leader designated as $j^* = N$. Consequently, a new leader must be appointed, and the compromised vehicle (previous leader) is reclassified as a follower. Under the updated communication topology, all vehicles are required to track the state of the newly designated leader, identified as vehicle $N - 1$. Accordingly, the following dynamics are formulated to represent the behavior of both the new leader and the previously attacked vehicle:

$$\dot{X}(t) = \bar{\Phi}X(t) + \bar{\Phi}_1X(t - \tau(t)) \quad (9)$$

where $\bar{\Phi} \in \mathbb{R}^{2N \times 2N}$ determines the impact of the state and also $\bar{\Phi}_1 \in \mathbb{R}^{2N \times 2N}$ denotes the influence of the delayed state, which is presented by:

$$\bar{\Phi} = \begin{bmatrix} 0_{N \times N} & I_{N \times N} \\ -\bar{\mathcal{L}}_{N \times N} & -\gamma \bar{\mathcal{L}}_{N \times N} \end{bmatrix}, \quad \bar{\Phi}_1 = \begin{bmatrix} 0_{N \times N} & 0_{N \times N} \\ -\bar{\mathcal{G}} & -\gamma \bar{\mathcal{G}} \end{bmatrix} \quad (10)$$

where $\bar{\mathcal{R}} = [\bar{r}_{ij}] \in \mathbb{R}^{N \times N}$ is regarded as $\bar{\mathcal{R}} = \mathcal{L}_1 - \bar{\mathcal{L}}$.

Hence, by decoupling the leader's state from the error dynamics described in Eq. (9), the following result is obtained:

$$\dot{\Psi}(t) = \tilde{\Phi}\Psi(t) + \tilde{\Phi}_1\Psi(t - \tau(t)) \quad (11)$$

Algorithm 1 DoS Attack Detection and Delay Measurement

Input: $x_i(t) = [\xi_i(t), v_i(t)]^T$, reference state $x_{i,\text{ref}}(t)$, initial timers $T_{i,1} = T_{i,2} = 0$

Output: $\tau_i(t)$: Delay measurement induced by a DoS attack

```

1: for  $i = 1, 2, \dots, N - 1$  do
2:   Receive current state  $x_i(t)$  and reference state  $x_{i,\text{ref}}(t)$ 
3:   if  $\|x_i(t) - x_{i,\text{ref}}(t)\| \leq \varepsilon$  then
4:     Set  $\tau_i(t) = 0$ ; and continue
5:   end if
6:   while  $\|x_i(t) - x_{i,\text{ref}}(t)\| > \varepsilon$  do
7:     Declare Vehicle  $i$  as isolated; stop Timer 1
8:     Store last reference:  $X_{i,r} = x_{i,\text{ref}}(t)$ 
9:     DoS attack detected on Vehicle  $i$ 
10:    while true do
11:      if  $x_i(t) = X_{i,r}$  then
12:        Stop Timer 2
13:      break
14:    end if
15:  end while
16:  Compute delay:  $\tau_i(t) = T_{i,2} - T_{i,1}$ ; reset both timers
17: end while
18: end for

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where $\Psi(t) = [P_z^T(t), V_z^T(t)]^T$ denotes the augmented error vector, with $P_z(t) = [\hat{\xi}_1(t), \hat{\xi}_2(t), \dots, \hat{\xi}_{N-1}(t)]^T$ representing the position error components, and $V_z(t) = [\hat{v}_1(t), \hat{v}_2(t), \dots, \hat{v}_{N-1}(t)]^T$ indicating the corresponding velocity error components.

The influence of the error signal in Eq. (11) is captured by the matrix $\tilde{\Phi} \in \mathbb{R}^{2(N-1) \times 2(N-1)}$, defined as:

$$\tilde{\Phi} = \begin{bmatrix} 0_{(N-1) \times (N-1)} & I_{(N-1) \times (N-1)} \\ -\mathcal{K} & -\gamma \mathcal{K} \end{bmatrix} \quad (12)$$

where $\mathcal{K} = [k_{ij}] \in \mathbb{R}^{(N-1) \times (N-1)}$ is computed as $k_{ij} = \hat{l}_{ij} - \hat{l}_{Nj}$. Additionally, $\tilde{\Phi}_1 \in \mathbb{R}^{2(N-1) \times 2(N-1)}$ characterizes the influence of the delayed error on the system, given by:

$$\tilde{\Phi}_1 = \begin{bmatrix} 0_{(N-1) \times (N-1)} & I_{(N-1) \times (N-1)} \\ -\mathcal{H} & -\gamma \mathcal{H} \end{bmatrix} \quad (13)$$

where $\mathcal{H} = [h_{ij}] \in \mathbb{R}^{(N-1) \times (N-1)}$ is achieved as $h_{ij} = g_{ij} - g_{Nj}$ [13].

Remark 3: The matrix $\mathcal{H}$ encapsulates all non-zero eigenvalues of the Laplacian matrix associated with Eq. (4).

III. DETECTION OF DOS ATTACKS

Two incremental timers, T1 and T2, are introduced in Algorithm 1 to detect vehicles attacked by DoS attacks and to estimate the injected delay $\tau_i(t)$. The algorithm operates on reference models defined in Eqs. (4)–(9), predicting the nominal behavior of each vehicle under attack-free conditions. Incorporating detection strategy shown in Algorithm 1 enables earlier identification of anomalies, allowing compromised vehicles to be isolated quickly and preventing corrupted data from spreading through the communication graph. This capability is particularly important in scenarios where both the leader and a follower are simultaneously

attacked, as timely detection supports the fast transition to a resilient topology.

Algorithm 1 implements detection through two counters assigned to each vehicle. Under normal conditions, Timer 1 and Timer 2 run simultaneously with equal values (line 4). When a vehicle becomes compromised, its state trajectory $x_i(t)$ deviates from the reference $x_{i,\text{ref}}(t)$ (line 6). At this point, Timer 1 is stopped, while Timer 2 continues until the actual state matches the last valid state stored in $X_{i,r}$ (line 11). The induced delay is calculated as the difference between the final values of the two timers, providing a quantitative measure of the attack's impact. This delay information is then used to guide the design of a resilient control strategy that adapts to both the timing and severity of detected attacks.

IV. RESILIENT CONTROL

Consider the closed-loop dynamics expressed in Eq. (4) and Eq. (9) for a consensus-based platoon with N vehicles. When the j^* th vehicle is compromised by a DoS attack, Algorithm 2 is employed to reintegrate the attacked vehicle into the leader-follower framework, with particular emphasis on cases where the leader is attacked.

Using the delay $\tau_i(t)$ estimated from Algorithm 1, the resilient controller generates control actions that respect inter-vehicle spacing and input limits while adapting to the timing and severity of attacks. This predictive structure enables safe reintegration of compromised vehicles and enhances resilience under adversarial conditions.

When the leader is attacked, it relinquishes its role and is reassigned as a follower. A new leader is then selected among the all members of the platoon according to section IV-A. Therefore, the selected follower becomes the root node in the updated spanning tree, thereby sustaining consensus and structural integrity.

Finally, as presented in Algorithm 2, the resilient controller ensures smooth topological reconfiguration by coordinating vehicle dynamics during the leadership transition, reducing transient errors, and preserving stability and cohesion across the platoon.

A. Data-Driven Similarity Analysis for Leader Reassignment

To enable a smooth and reliable leader reassignment under DoS attacks, a data-driven similarity analysis is employed to evaluate candidate vehicles in the vicinity of the compromised leader. The objective of this analysis is to identify a vehicle whose motion behavior is most consistent with the nominal leader dynamics, thereby minimizing transient disturbances during topology reconfiguration.

The proposed similarity analysis relies on short-horizon time-series measurements of vehicle positions and velocities, which are directly available through onboard sensing and vehicle-to-vehicle communication. Unlike learning-based approaches, the method does not require offline training, historical datasets, or model identification, making it suitable for real-time implementation in practical platooning systems.

Algorithm 2 Resilient Consensus Under DoS Attacks

Input: $x_i(t) = [\xi_i(t), v_i(t)]^T$, reference trajectory, $\mathcal{B}$: Critical area upper bound, $\tau_i(t)$ from Algorithm 1

Output: $v_i(t)$

```

1: while true do
2:   for  $i = 1, 2, \dots, N-1$  do
3:     if  $\tau_i(t) < \mathcal{B}$  then
4:       Enable switching signal for vehicle  $i$ 
5:       Determine  $v_i(t)$ 
6:     else
7:       Set  $v_i(t) \rightarrow 0$  to stop vehicle  $i$ 
8:     end if
9:   end for
10: end while

```

For each candidate vehicle, motion consistency is quantified by comparing its recent velocity profile with that of the former leader over a finite time window $[t-T, t]$. Specifically, a velocity similarity metric is defined as

$$S_i = \frac{1}{T} \int_{t-T}^t \|v_i(\tau) - v_\ell(\tau)\|^2 d\tau, \quad (14)$$

where $v_i(\tau)$ denotes the velocity of the i -th candidate vehicle and $v_\ell(\tau)$ represents the velocity of the former leader. A smaller value of S_i indicates a higher degree of motion similarity and a greater suitability for leader reassignment.

In addition to velocity consistency, spatial proximity to the former leader and communication connectivity are considered to ensure minimal disruption to the platoon structure. These criteria are combined within a rule-based multi-criteria decision strategy to select the new leader. By leveraging real-time time-series data and lightweight similarity metrics, the proposed approach enables explainable and computationally efficient leader reassignment while preserving platoon stability under DoS attacks.

Remark 4: In practice, attacked vehicles decelerate before stopping. Likewise, when a DoS attack is detected and the delay exceeds the critical value $\mathcal{B}$, the attacked vehicle's acceleration decreases ($\dot{v}_i(t) < 0$) until its velocity approaches zero ($v_i(t) \rightarrow 0$). As detailed in Algorithm 1, the compromised vehicle and its associated delay are identified using dual incremental counters. Subsequently, Algorithm 2 executes the recovery strategy.

A resilient vehicular control strategy is implemented within a switching system framework to recover deviated vehicles when the DoS-induced delay satisfies the critical condition $\tau(t) < \mathcal{B}$. Here, $\mathcal{B}$ is a design value bounded by the upper limit U defined in Eq. (6). The pivotal region is dynamically adjusted based on the network's sensitivity, determining whether an attacked vehicle should be recovered or isolated. As described in line 3 of Algorithm 2, if an vehicle's measured delay falls below the lower boundary of this region, the switching mechanism activates to reintegrate the vehicle into the leader-follower consensus framework.

Definition 2 (see [21]): Consider a platoon of intelligent vehicles with homogeneous dynamics, each governed by

the distributed controller defined in Eq. (2) and modeled by the dynamics in Eq. (1). Suppose that the switching signal operates on three distinct modes, determined according to the following switching law:

$$\mathcal{N}_{\sigma_i}(t_k, t_{k+1}) \leq \mathcal{N}_{0i} + \frac{\mathcal{T}_i(t_k, t_{k+1})}{\tau_{ai}} \quad (15)$$

where τ_{ai} denotes the mode-dependent average dwell time (MDADT), $\mathcal{T}_i(t_k, t_{k+1})$ is the duration of interval $[t_k, t_{k+1})$, $\mathcal{N}_{0i}$ is the chatter bound, and $\mathcal{N}_{\sigma_i}(t_k, t_{k+1})$ represents the number of switching events, the system ensures recovery and reintegration of the compromised vehicle into the leader-follower consensus.

Remark 5: The switching strategy follows Eq. (15) and employs MDADT τ_a to maintain consensus stability. Properly selected switching intervals guarantee exponential convergence and asymptotic stability by ensuring the Lyapunov-Krasovskii derivative remains negative definite over each interval.

If the measured delay exceeds a critical threshold, the switching logic halts the compromised vehicle, as outlined in line 7 of Algorithm 2.

V. STABILITY ANALYSIS

Theorem 5.1: The system described by Eq. (11), subject to the bounded delay condition in Eq. (6), is asymptotically stable if and only if the following condition holds:

$$\begin{bmatrix} \psi_{11} & \psi_{12} & R \\ \psi_{21} & \psi_{22} & 0 \\ * & * & -R \end{bmatrix} < 0$$

where

$$\psi_{11} = \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi} - R + S + \tilde{\Phi}^T \mathcal{W}_p + \mathcal{W}_p \tilde{\Phi} \quad (16)$$

$$\psi_{12} = \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi}_1 + \mathcal{W}_p \tilde{\Phi}_1 \quad (17)$$

$$\psi_{21} = \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi} + \tilde{\Phi}_1^T \mathcal{W}_p \quad (18)$$

$$\psi_{22} = \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi}_1 - (1-d)S \quad (19)$$

where $R \in \mathbb{R}^{2(N-1) \times 2(N-1)}$, $S \in \mathbb{R}^{2(N-1) \times 2(N-1)}$ and $\mathcal{W}_p \in \mathbb{R}^{2(N-1) \times 2(N-1)}$ are all positive definite matrices. As specified in Eq. (6), the parameters $\mathbf{U}$ and d represent the upper bounds on the communication delay and its time derivative, respectively.

Proof: Consider the following Lyapunov-Krasovskii functional, which is composed of three components:

$$\begin{aligned} V(t, \Psi(t), \dot{\Psi}(t)) = & \mathbf{U} \int_{-U}^0 \int_{t+\beta}^t \dot{\Psi}^T(s) R \dot{\Psi}(s) ds d\beta \\ & + \int_{t-\tau(t)}^t \Psi^T(s) S \Psi(s) ds + \Psi^T(t) \mathcal{W}_p \Psi(t) \end{aligned} \quad (20)$$

where the first term captures the maximum delay, the second represents the time-varying delay, and the third, independent of delay, describes the switched subsystem dynamics essential for ensuring system stability.

The Lyapunov-Krasovskii functional term V_1 is defined as

$$V_1(t) = \mathbf{U} \int_{t-U}^t (s-t+\mathbf{U}) \dot{\Psi}^T(s) R \dot{\Psi}(s) ds. \quad (21)$$

Taking the time derivative yields

$$\dot{V}_1 = \mathbf{U}^2 \dot{\Psi}^T(t) R \dot{\Psi}(t) - \mathbf{U} \int_{t-U}^t \dot{\Psi}^T(s) R \dot{\Psi}(s) ds, \quad (22)$$

which is decomposed as $\dot{V}_1 = \dot{V}_{11} + \dot{V}_{12}$.

Substituting the system dynamics (11) into $\dot{V}_{11}$ gives

$$\begin{aligned} \dot{V}_{11} = & \mathbf{U}^2 \left(\Psi^T(t) \tilde{\Phi} + \Psi^T(t-\tau(t)) \tilde{\Phi}_1^T \right) \cdot R \\ & \cdot \left(\tilde{\Phi} \Psi(t) + \tilde{\Phi}_1 \Psi(t-\tau(t)) \right) \end{aligned} \quad (23)$$

Define the augmented error vector

$$\zeta(t) = [\Psi^T(t) \quad \Psi^T(t-\tau(t)) \quad \Psi^T(t-U)]^T \quad (24)$$

Then, $\dot{V}_{11}$ can be compactly expressed as

$$\dot{V}_{11} = \zeta^T(t) \begin{bmatrix} \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi} & \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi}_1 & 0 \\ \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi} & \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi}_1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \zeta(t) \quad (25)$$

which is suitable for subsequent LMI formulation.

With respect to Eq. (21), taking the derivative of the second term of Eq. (22) yields:

$$\dot{V}_{12} = -\mathbf{U} \int_{t-U}^t \dot{\Psi}^T(s) R \dot{\Psi}(s) ds \quad (26)$$

Subsequently, by applying Jensen's inequality [22], the expression in Eq. (26) can be simplified as:

$$\dot{V}_{12} \leq -\left(\int_{t-U}^t \dot{\Psi}^T(s) ds \right) R \left(\int_{t-U}^t \dot{\Psi}(s) ds \right) \quad (27)$$

$$\dot{V}_{12} \leq (-\Psi^T(t) + \Psi^T(t-U)) R (\Psi(t) - \Psi(t-U)) \quad (28)$$

$$\dot{V}_{12} \leq \zeta^T(t) \begin{bmatrix} -R & 0 & R \\ 0 & 0 & 0 \\ R & 0 & -R \end{bmatrix} \zeta(t) \quad (29)$$

Based on Eq. (25) and Eq. (29), the linear matrix inequality (LMI) associated with the first component of the Lyapunov function in Eq. (21) is given by:

$$\dot{V}_1 \leq \zeta^T(t) \begin{bmatrix} \phi_{11} & \phi_{12} & R \\ \phi_{21} & \phi_{22} & 0 \\ * & * & -R \end{bmatrix} \zeta(t) \quad (30)$$

where

$$\phi_{11} = \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi} - R; \quad \phi_{12} = \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi}_1$$

$$\phi_{21} = \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi}; \quad \phi_{22} = \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi}_1$$

Subsequently, the second term of Eq. (20) can be expressed as:

$$V_2(t, \Psi(t)) = \int_{t-\tau(t)}^t \Psi^T(s) S \Psi(s) ds \quad (31)$$

Differentiating the aforementioned function results in:

$$\dot{V}_2 = \Psi^T(t) S \Psi(t) - (1-\dot{\tau}(t)) \Psi^T(t-\tau(t)) S \Psi(t-\tau(t)) \quad (32)$$

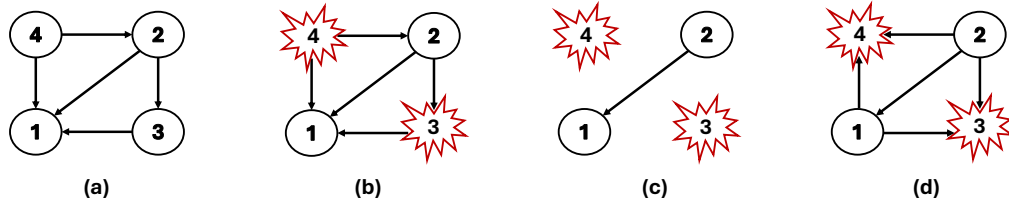


Fig. 1: Communication graph diagrams for a 4-vehicle platoon. (a) Normal operation with no attack. (b) The original leader (Vehicle 4) and a follower (Vehicle 3) are under DoS attack and successfully detected. (c) Isolation of the compromised vehicles from the communication network. (d) Vehicle 2 assumes the leadership role and reestablishes communication with the remaining vehicles, restoring the leader–follower consensus within the platoon.

Given that $\hat{\tau}(t) < d$ as indicated in Eq. (6), the resulting expression can be formulated as:

$$\dot{V}_2 \leq \Psi^T(t) S \Psi(t) - (1-d) \Psi^T(t-\tau(t)) S \Psi(t-\tau(t)) \quad (33)$$

By substituting Eq. (24) into the preceding inequality, the expression can be reformulated as the following linear matrix inequality (LMI):

$$\dot{V}_2 \leq \zeta^T(t) \begin{bmatrix} S & 0 & 0 \\ 0 & -(1-d)S & 0 \\ 0 & 0 & 0 \end{bmatrix} \zeta(t) \quad (34)$$

Consequently, the third component of the Lyapunov function in Eq. (20) is given by:

$$V_3(t, \Psi(t)) = \Psi^T(t) \mathcal{W}_p \Psi(t) \quad (35)$$

where $\mathcal{W}_p \succ 0$ is a positive definite matrix and $\sigma(t) \in K = \{1, 2, \dots, S\}$ denotes the switching signal over S subsystems. A mode transition occurs at t_i when $\sigma(t_i^-) = f$ and $\sigma(t_i) = p$, with $p \neq f$. Differentiating (35) and using the dynamics in (11) yields:

$$\begin{aligned} \dot{V}_3 &= \text{sym}(\dot{\Psi}^T(t) \mathcal{W}_p \Psi(t)) \\ &= \dot{\Psi}^T(t) \mathcal{W}_p \Psi(t) + \Psi^T(t) \mathcal{W}_p \dot{\Psi}(t) \\ &= (\Psi^T(t) \tilde{\Phi}^T + \Psi^T(t-\tau(t)) \tilde{\Phi}_1^T) \mathcal{W}_p \Psi(t) \\ &\quad + \Psi^T(t) \mathcal{W}_p (\tilde{\Phi} \Psi(t) + \tilde{\Phi}_1 \Psi(t-\tau(t))) \\ &= \Psi^T(t) (\tilde{\Phi}^T \mathcal{W}_p + \mathcal{W}_p \tilde{\Phi}) \Psi(t) \\ &\quad + \Psi^T(t) \mathcal{W}_p + \Psi^T(t-\tau(t)) \tilde{\Phi}_1^T \mathcal{W}_p \Psi(t) \end{aligned} \quad (36)$$

where, in accordance with Eq. (15), if the previously simplified expression is represented as Eq. (30) and Eq. (34), the corresponding linear matrix inequality (LMI) can be expressed as:

$$\dot{V}_3 = \zeta^T(t) \begin{bmatrix} \tilde{\Phi}^T \mathcal{W}_p + \mathcal{W}_p \tilde{\Phi} & \mathcal{W}_p \tilde{\Phi}_1 & 0 \\ \tilde{\Phi}_1^T \mathcal{W}_p & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \zeta(t) \quad (37)$$

Finally, based on Eq. (30), Eq. (34), and Eq. (37), the complete linear matrix inequality (LMI) associated with the Lyapunov function in Eq. (20) is formulated as:

$$\begin{aligned} \dot{V} &= \dot{V}_1 + \dot{V}_2 + \dot{V}_3 \\ &\leq \zeta^T(t) \begin{bmatrix} \psi_{11} & \psi_{12} & R \\ \psi_{21} & \psi_{22} & 0 \\ * & * & -R \end{bmatrix} \zeta(t) \end{aligned} \quad (38)$$

where

$$\begin{aligned} \psi_{11} &= \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi} - R + S + \tilde{\Phi}^T \mathcal{W}_p + \mathcal{W}_p \tilde{\Phi} \\ \psi_{12} &= \mathbf{U}^2 \tilde{\Phi}^T R \tilde{\Phi}_1 + \mathcal{W}_p \tilde{\Phi}_1 \\ \psi_{21} &= \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi} + \tilde{\Phi}_1^T \mathcal{W}_p \\ \psi_{22} &= \mathbf{U}^2 \tilde{\Phi}_1^T R \tilde{\Phi}_1 - (1-d)S \end{aligned}$$

Now, based on Eq. (38), if the following matrix is negative definite, it ensures the asymptotic stability of the system described by Eq. (11), thereby concluding the proof.

$$\begin{bmatrix} \psi_{11} & \psi_{12} & R \\ \psi_{21} & \psi_{22} & 0 \\ * & * & -R \end{bmatrix} < 0 \quad (39)$$

Remark 6: It is essential to note that the incremental timers employed for measuring the delay must be reset to release the previously computed delay values, thereby allowing the proposed control strategy to be applicable across various operational scenarios.

VI. SIMULATION RESULTS

Consider the dynamics of each vehicle in the platoon as described by Eq. (1), which incorporates a stabilizing local feedback controller:

$$\dot{x}_i(t) = \begin{bmatrix} 0 & 1 \\ -8 & -7 \end{bmatrix} x_i(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \hat{u}_i(t) \quad (40)$$

In the communication topology shown in Fig. 1(a), vehicle 4 serves as the platoon leader. Fig. 2 displays the position and velocity trajectories under nominal (attack-free) conditions.

We consider a scenario in which the 4-th vehicle, serving as the platoon leader, and the 3-th vehicle, a follower, are simultaneously subjected to a DoS attack, as shown in Fig. 1(b). In the first phase, the compromised vehicles are detected using the DoS attack identification strategy combined with two incremental timers described in Algorithm 1. This mechanism enables estimation of the delay for each attacked vehicle.

Remark 7: The communication delay, representing the strength of the DoS attack in the simulation, is modeled as a nonlinear, time-varying signal rather than a constant or sinusoidal profile, reflecting realistic DoS attack dynamics.

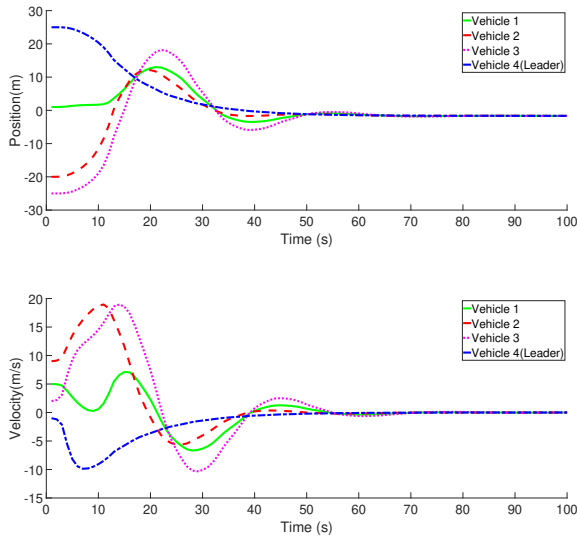


Fig. 2: Vehicle positions (top) and velocities (bottom) under normal conditions with Vehicle 4 acting as the leader

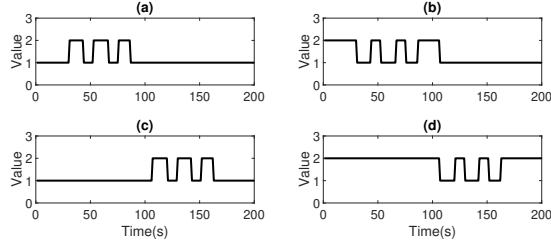


Fig. 3: Switching signals triggered in response to detected DoS attacks.

In the second phase, once the 4-th vehicle is confirmed under attack, it relinquishes its leadership and transitions to a follower in the reconfigured communication graph. Vehicle 2 is then appointed as the new leader and designated as the root node in the updated spanning tree (Fig. 1(c)). The selection follows three criteria: (i) Vehicle 2 has more outward communication links to disseminate its state effectively, (ii) it is closest to the former leader (Vehicle 4), and (iii) its velocity profile closely matches that of the former leader at the time of detection. This careful selection minimizes disruption and preserves platoon stability.

In the third phase, the former leader is reassigned as a follower through Algorithm 2, while the compromised Vehicles 3 and 4 are isolated and later reintegrated into the leader-follower consensus based on the delay estimated by Algorithm 1. The resilient controller enables timely deviation assessment and appropriate recovery actions according to the switching signals in Fig. 3, with a critical delay threshold of 15 s. Without DoS attacks, consensus is achieved according to Eqs. (4), (6), and (11) under the communication topology in Fig. 1. When Vehicles 3 and 4 are attacked, their delays are computed; if a delay remains below the threshold, the corresponding vehicle is isolated using the switching signals in Figs. 3(a) and 3(b). The resulting position and velocity trajectories are shown in Fig. 4.

The communication graph with isolated Vehicles 4 and 3

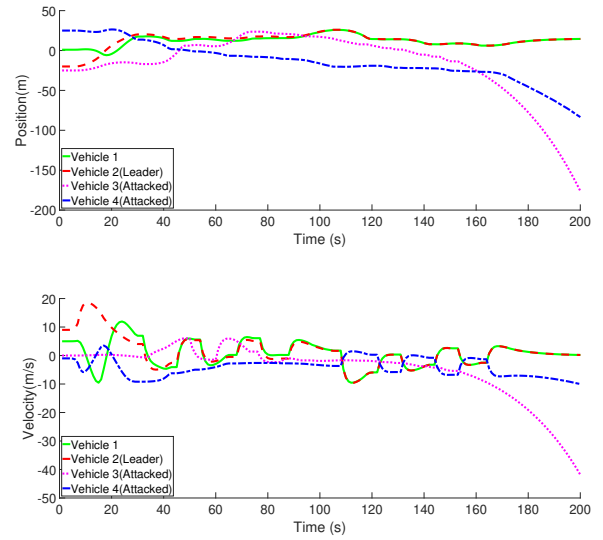


Fig. 4: Vehicle positions (top) and velocities (bottom) under DoS attacks, with respect to inter-vehicle spacing. Vehicle 2 serves as the leader.

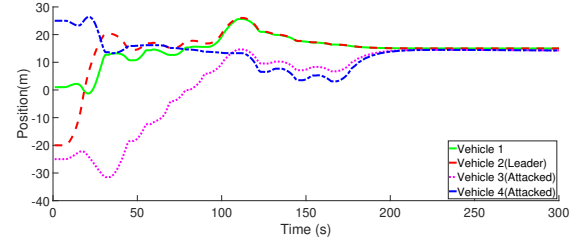


Fig. 5: Positions of the vehicles, where Vehicles 3 and 4 are under attack. Vehicle 2 assumes the role of the new leader.

is shown in Fig. 1(c). After the simultaneous DoS attacks on Vehicle 4, the former leader, and Vehicle 3, a follower, are detected at $t = 30$ s using Algorithm 1, the resilient strategy in Algorithm 2 is activated. Since the injected delay is time-varying, the measured delay is continuously monitored and updated. When the delay remains below the 15 s threshold, the resilient controller operates with the switching protocols in Figs. 3(c) and 3(d). During recovery, the attacked vehicles are reintegrated into the platoon under communication constraints. They can receive information from the new leader and neighboring normal vehicles identified by the detection algorithm, but cannot transmit information to neighbors, as shown in Fig. 1(d). This selective reception improves consensus recovery and enables the treated vehicles to track the leader state again. Consequently, the switching mechanism becomes inactive, as shown in Fig. 5. The position and velocity responses in Fig. 6 confirm platoon recovery while satisfying the spacing constraints $d_{1,2} = 4$ m, $d_{3,2} = 10$ m, and $d_{4,2} = 20$ m.

The switching signal alternates between two subsystems: the configuration in Fig. 1(b) and the updated topology in Fig. 1(c) representing the recovery process. By dynamically transitioning between these subsystems through the designed switching mechanism, the system successfully restores con-

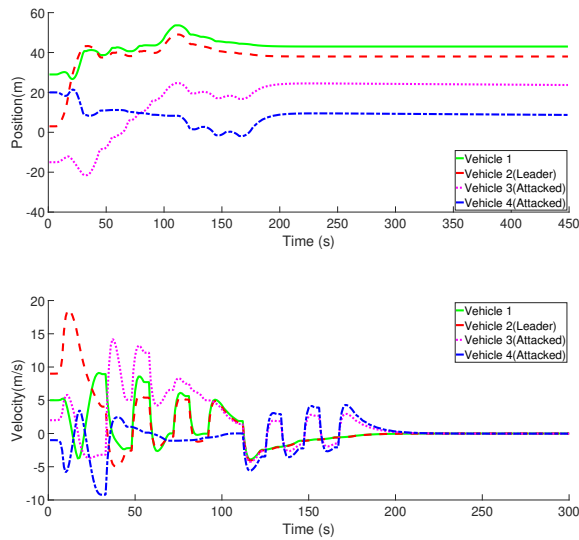


Fig. 6: Vehicle positions (top) and velocities (bottom) under DoS attacks, with respect to inter-vehicle spacing. Vehicle 2 is designated as the leader.

sensus, as confirmed by the simulation results.

The motion-similarity term used for leader reassignment is computed over a short sliding window of length T using sampled position and velocity data. In the simulations, signals are sampled with period $T_s = 0.1s$, and the similarity window is set to $T = MT_s$, where $M = 50$ is the number of samples in the window. All similarity measures are evaluated relative to the former leader's recent motion profile.

VII. CONCLUSION

This paper presented a resilient vehicular control framework to ensure safe platooning under DoS attacks, particularly when both the leader and a follower are compromised simultaneously. DoS attack detection is achieved by comparing actual vehicle states with reference states, supported by dual incremental timers to estimate time-varying delays (varying strength of the DoS attack) and identify attacked vehicles. Upon detection, the network reconfigures: the compromised leader is demoted, a new leader is assigned, and the attacked vehicles are driven back to consensus via a distributed resilient controller, minimizing disruption and enabling rapid recovery.

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